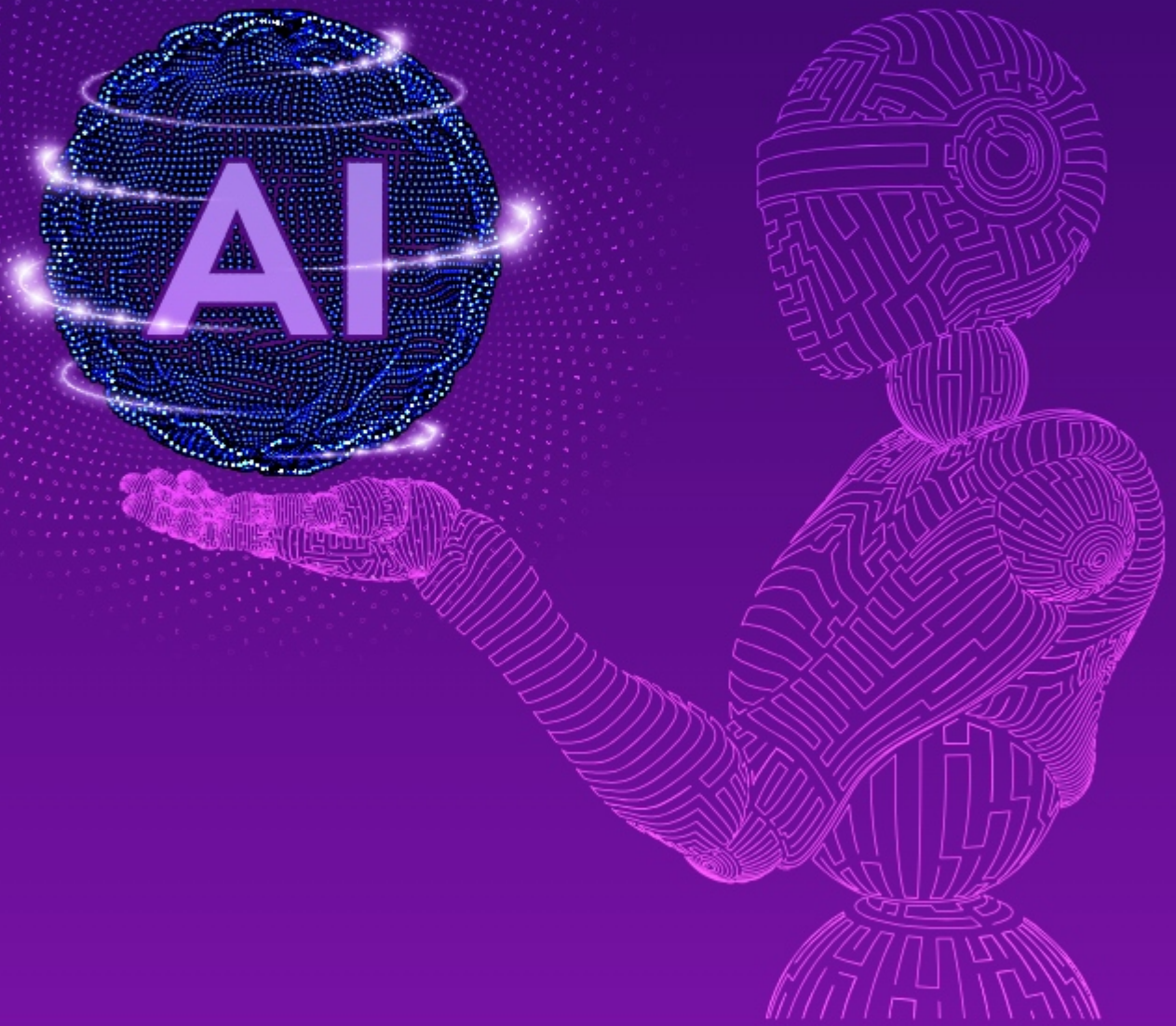




GeeksforGeeks

GENRATIVE AI

TRAINING PROGRAM



~ Detailed Course Syllabus ~



SESSIONS

Session-1

Python Basics for Data Science

- Learn Python basics focused on data analysis and AI use cases.
- Work with core syntax, variables, data types, and control flow (if, loops).
- Write reusable code using functions and organize logic into modules.
- Build a basic understanding of Object-Oriented Programming (OOP) to prepare for model and project code.

Session-2

Feedforward Neural Networks

- Understand neurons, layers, and activation functions are in a neural network.
- Walk through the forward pass: how input data flows through layers to produce predictions.
- Learn the idea of loss functions and optimization at a conceptual level.
- Build your first simple feedforward neural network and see it learn from data.

Session-3

Backpropagation & Digit Classifier Project

- Study the backpropagation algorithm and how gradients are used to update weights.
- Understand the role of the learning rate in how fast or slow a model learns.
- Get an intuitive feel for vanishing and exploding gradients.
- Project: Implement and train a simple neural network for digit classification end-to-end.



SESSIONS

Session-4

Convolutional Neural Networks (CNNs)

- Learn how convolutions, filters, and feature maps extract patterns from images.
- Use pooling layers like max pooling and average pooling to reduce spatial dimensions.
- Explore typical CNN architectures used for image tasks and their building blocks.

Session-5

Recurrent Neural Networks (RNNs)

- Understand sequence modeling basics for time-ordered data such as text or time-series.
- See how vanilla RNNs process sequences step by step and where their limitations appear.
- Look at common use cases of RNNs: text modeling, time-series forecasting, and sentiment analysis.

Session-6

Backpropagation & Digit Classifier Project

- Learn the intuition behind self-attention and why it powers modern LLMs.
- Understand the idea of positional encoding and the high-level structure of encoder and decoder.
- Project: Combine earlier concepts to build a Digit Classifier + IMDB Sentiment Classification pipeline.

SESSIONS

Session-7

Hugging Face Essentials

- Get an overview of the Hugging Face Hub and how models are shared.
- Learn how to load pre-trained models and tokenizers in a few lines of code.
- Use pipelines for quick inference across text, vision, and audio without writing low-level model code.

Session-8

Gen AI & LangChain Workflows

- Learn the key terms and concepts used in Generative AI and LLM discussions.
- Understand LangChain basics: chains, tools, and memory for building LLM flows.
- Orchestrate multi-step LLM workflows that combine prompts, tools, and responses.

Session-9

Prompt Engineering & Mini-Projects

- Study common prompt patterns and practical best practices.
- Use few-shot and chain-of-thought style prompts to improve model outputs.
- **Project:** Build multiple mini-projects using pipelines (sentiment, text generation, QA, translation, NER, simple chatbot, and more small tasks).

SESSIONS

Session-10

Embeddings & Vector Databases

- Understand what embeddings are and how they turn text into vectors.
- Learn how text/vector similarity search works conceptually.
- Get introduced to vector databases and perform basic similarity queries.

Session-11

Retrieval-Augmented Generation (RAG)

- See how RAG architecture combines retrieval with generation in a single flow.
- Learn document chunking and indexing for better search over your data.
- Build applications that answer questions using private or custom documents instead of just the base model.

Session-12

Transfer Learning with QLoRA & PEFT

- Compare full fine-tuning with parameter-efficient tuning approaches.
- Understand the basic ideas behind QLoRA and PEFT for adapting large models.
- Project: Build and run a RAG-based chatbot on your own dataset.

SESSIONS

Session-13

Diffusion Models

- Learn the basics of generative modeling for images.
- Understand the diffusion and denoising process step by step.
- Explore common image generation use cases and how diffusion models fit there.

Session-14

Multi-Modal Gen AI & Dev Setup

- Learn what text-image and multi-modal models are and how they work conceptually.
- Use models that can work with both text and images in the same pipeline.
- Set up basic shell and Git workflows for managing AI projects from the terminal.

Session-15

Creative AI Project

- Apply version control best practices with Git to your AI project code.
- Project: Build a newsletter or creative AI content system (blogs, social posts, visuals with AI support) as a full mini-project.

SESSIONS

Session-16

FastAPI & Docker for AI APIs

- Learn how to build REST APIs with FastAPI to serve AI models.
- Expose ML/LLM models via HTTP endpoints for use in apps or services.
- Containerize your application with Docker to prepare it for deployment.

Session-17

Local LLMs in Practice

- Learn the basics of running Large Language Models on local hardware.
- Understand model selection and the idea of quantization.
- Manage resources and constraints (RAM, VRAM, CPU) when working with local LLMs.

Session-18

Streamlit & Gradio UI for Models

- Build interactive UIs for your models using Streamlit.
- Create quick visual demos and playgrounds using Gradio.
- **Project:** Deploy and build a user interface for a RAG chatbot.

SESSIONS

Session-19 Introduction to Agentic AI

- Understand the difference between simple LLM calls and agents that take actions.
- Learn how tools, actions, and planning fit into agent workflows.
- Explore common use cases for autonomous AI agents in practical setups.

Session-20 CrewAI & Multi-Agent Systems

- Get an overview of CrewAI or similar multi-agent frameworks.
- Design multi-agent workflows where different agents play different roles.
- **Project:** Build a market research or autonomous AI agent system.

Session-21 LLM Guardrails & Safety

- Learn prompt-level and policy-level guardrails to control outputs.
- Implement content filtering and basic safety checks.
- Use these ideas to start building responsible AI applications.

SESSIONS

Session-22 **Embeddings & Vector Databases**

- Learn quantitative and qualitative ways to evaluate LLM outputs.
- Work with metrics such as accuracy and BLEU/ROUGE-style scores, plus hallucination checks.
- Apply human-in-the-loop evaluation patterns to refine your systems.

Session-23 **Retrieval-Augmented Generation (RAG)**

- Define your capstone problem and its scope clearly.
- Select models, tools, and stack that match your idea.
- Draft the overall architecture and break work into concrete tasks.

Session-24 **Transfer Learning with QLoRA & PEFT**

- Do final integration and polish of your capstone system.
- Project: Present your final AI Capstone Showcase as a complete end-to-end solution.
- Walk through what you built, how it works, and what problem it solves.